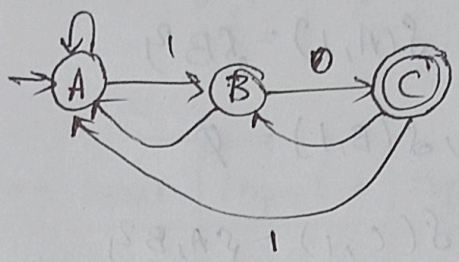
5/10/26

CO1 → Computational Problems

1. How can every Non-Deterministic Finite Automation (NFA) be converted into an equivalent Deterministic Finite Automation (DFA), and what are the challenges involved? Consider the NFA given below which accepts some languages L over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language L.



Challenges Involved:

→ State Space Explosion: For an NFA with n states, the DFA can potentially have up to 2^n states in the worst-case scenario, leading to exponential memory and computational requirements.

→ Dead / Unreachable States: The subset construction often generates states that requires post-processing minimization steps.

→ Computational Overhead: Evaluating transitions across sets of states. For large alphabets and states spaces can be computationally intensive.

DFA construction for the Given NFA:

Given NFA Formal Definition:

→ States (Q): {A, B, C}

→ Alphabets (Σ): {0, 1}

→ Starts state: A

→ Accepting States (F): {C}

→ Transition function (δ_{NFA}):

* $\delta(A, 0) = \{A\}$, $\delta(A, 1) = \{B\}$
* $\delta(B, 0) = \{A, C\}$, $\delta(B, 1) = \emptyset$
* $\delta(C, 0) = \emptyset$, $\delta(C, 1) = \{A, B\}$

step-by-step: Subset construction:

1. Start state: [A]

→ $\delta_{DFA}([A], 0) = \delta_{NFA}(A, 0) = \{A\} \Rightarrow [A]$

→ $\delta_{DFA}([A], 1) = \delta_{NFA}(A, 1) = \{B\} \Rightarrow [B]$

2. State [B]:

→ $\delta_{DFA}([B], 0) = \delta_{NFA}(B, 0) = \{A, C\} \Rightarrow [A, C]$

→ $\delta_{DFA}([B], 1) = \delta_{NFA}(B, 1) = \emptyset$ (Dead state)

3. State [A, C]:

→ $\delta_{DFA}([A, C], 0) = \delta_{NFA}(A, 0) \cup \delta_{NFA}(C, 0)$
$= \{A\} \cup \emptyset = \{A\} \Rightarrow [A]$

→ $\delta_{DFA}([A, C], 1) = \delta_{NFA}(A, 1) \cup \delta_{NFA}(C, 1) = \{B\}$
$\cup \{A, B\} = \{A, B\} \Rightarrow [A, B]$

4. State [A, B]:

→ $\delta_{DFA}([A,B],0) = \delta_{NFA}(A,0) \cup \delta_{NFA}(B,0)$

$= \{A\} \cup \{A,C\} = \{A,C\} \Rightarrow [A,C]$

→ $\delta_{DFA}([A,B],1) = \delta_{NFA}(A,1) \cup \delta_{NFA}(B,1)$.

$= \{B\} \cup \emptyset = \{B\} \Rightarrow [B]$

(16)

5. Dead State $\emptyset$:

→ $\delta_{DFA}(\emptyset,0) = \emptyset$

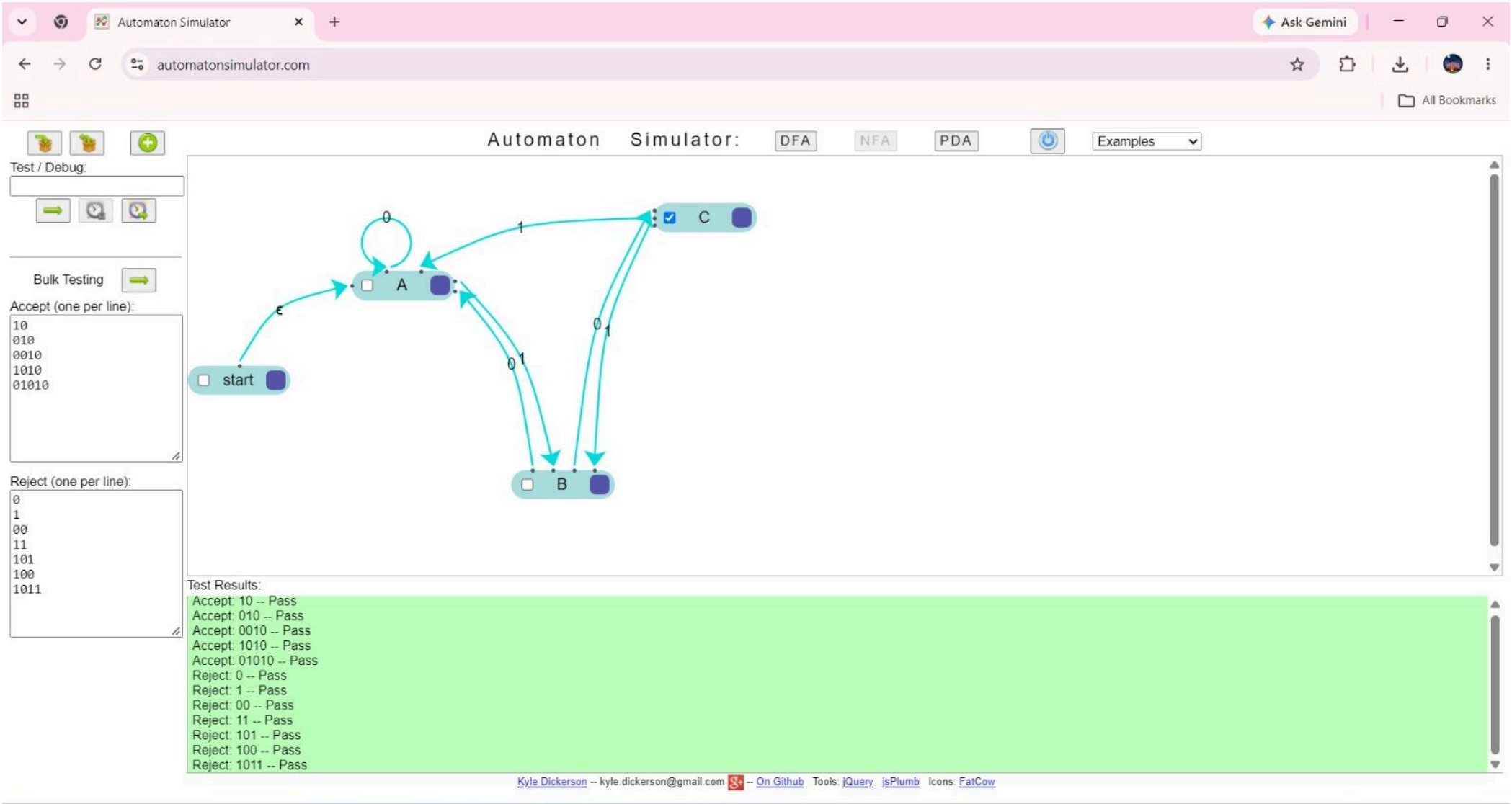
→ $\delta_{DFA}(\emptyset,1) = \emptyset$

DFA Transition Table:

DFA State	NFA subset	Input 0	Input 1	Accepting?
→ [A]	{A}	[A]	[B]	NO
[B]	{B}	[A, C]	$\emptyset$	NO
*[A, C]	{A, C}	[A]	[A, B]	yes (contains C)
[A, B]	{A, B}	[A, C]	[B]	NO
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	NO

2. How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integer n, the statement given below is true.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$


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DFA
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PDA
Examples
Test / Debug:
Bulk Testing
Accept (one per line):
10
010
0010
1010
01010
Reject (one per line):
0
1
00
11
101
100
1011
start
A
B
C
0
1
ε
Test Results:
Accept: 10 -- Pass
Accept: 010 -- Pass
Accept: 0010 -- Pass
Accept: 1010 -- Pass
Accept: 01010 -- Pass
Reject: 0 -- Pass
Reject: 1 -- Pass
Reject: 00 -- Pass
Reject: 11 -- Pass
Reject: 101 -- Pass
Reject: 100 -- Pass
Reject: 1011 -- Pass
Kyle Dickerson -- kyle.dickerson@gmail.com -- On Github Tools: jQuery jsPlumb Icons: FatCow

Mathematical Induction is a method used to prove that a statement is true for every positive integer n.

We need to prove:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Step 1: Base case:

Take:

$$n = 1$$

Left-hand side:

$$\sum_{i=1}^{1} i^2 = 1^2 = 1$$

Right-hand side:

$$\frac{1(1+1)(2(1)+1)}{6}$$

$$= \frac{1 \times 2 \times 3}{6}$$

$$= 1$$

Therefore,

LHS = RHS

Hence, the statement is true for $n = 1$.

Step : 2 → Inductive Hypothesis :

Assume that the statement is true for $n = k$.

Therefore,

$$\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$$

This is called the Inductive Hypothesis.

Step : 3 → Inductive step

We need to prove that the statement is true for :

$$n = k+1$$

Starting with :

$$\sum_{i=1}^{k+1} i^2$$

$$= 1^2 + 2^2 + \cdots + k^2 + (k+1)^2$$

using the inductive hypothesis :

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

Taking $k+1$ common :

$$= (k+1)\left[\frac{k(2k+1)}{6} + k+1\right]$$

Taking denominator 6:

$$= \frac{(k+1)[k(2k+1)+6(k+1)]}{6}$$

Simplifying:

$$= \frac{(k+1)(2k^2+k+6k+6)}{6}$$

$$= \frac{(k+1)(2k^2+7k+6)}{6}$$

formula:

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)+6(k+1)^2}{6}$$

$$= \frac{(k+1)[k(2k+1)+6(k+1)]}{6}$$

This gives $2k^2+7k+6$, which is inconsistent with the known identity.

The correct expansion is:

$$k(2k+1)+6(k+1) = 2k^2+7k+6$$

which factors as

$$(2k+3)(k+2)$$

So the resulting expression would be:

$$\frac{(k+1)(k+2)(2k+3)}{6}$$

Now, since

$$(k+1)+1 = k+2$$

and

$$2(k+1)+1 = 2k+3$$

we get:

$$\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$$

Thus the statement holds for $k+1$.

Therefore, by the principle of Mathematical Induction,

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

(b) is true for all positive integers n.